

PROBLEM STATEMENT

Failure to Thrive (FTT) is a complex pediatric condition impacting physical and emotional development.
Many parents and pediatricians lack accessible, personalized tools to manage and monitor it.
Existing solutions are fragmented, non-intuitive, and not tailored to each child's needs.

OUR SOLUTION

Personalized Care:

FTT treatment tailored to each child based on age, gender, and growth history. Covers nutrition, sleep, activity, and hydration.

Smart Plan Generation:

Automated creation of treatment plans using clinical logic and AI.

Data Tracking:

Web-based growth tracking using percentiles and dynamic charts. Enables easy updates, monitoring, and trend analysis.

SCIENTIFIC & CLINICAL BASIS

- All medical calculations and nutritional guidelines are based on World Health Organization (WHO) growth standards and dietary references.
- Growth percentiles and median weights are derived from internationally accepted WHO child growth charts.
- Macronutrient and micronutrient requirements are aligned with evidence-based pediatric nutrition literature.
- Lifestyle recommendations (hydration, sleep, physical activity) are derived from a comprehensive literature review of current pediatric health research.

ALGORITHM

Data Entry & Validation:

Users enter child's name, gender, birthdate, height, and weight. Inputs are validated for accuracy and clinical logic.

Growth History Generation:

A timeline with 6-month intervals is auto-generated from birth to present to track growth patterns.

Protocol Calculation:

Median weight is retrieved by age and gender. Calculates personalized calorie needs, macronutrients, and micronutrient goals (e.g., calcium, iron, vitamins).

Protocol Creation:

After completing all calculations, a personalized protocol is constructed—combining nutritional targets, hydration, activity, and sleep guidelines.

AI Treatment Plan:

The protocol is passed to an AI model that generates a full daily plan in Markdown: meals, hydration, activity, and sleep.

Daily Meal Plan ^

Update Meal Plan

Breakfast

Oatmeal with banana and honey

- 50 g oatmeal
- 1 medium banana
- 1 tsp honey

How to Prepare:

Cook oatmeal according to package instructions, slice banana on top, and drizzle with honey.

250 kcal 55 g carbohydrates
6 g protein 4 g fat

Lunch

Grilled chicken breast with steamed broccoli and brown rice

- 100 g chicken
- 100 g broccoli
- 75 g brown rice

How to Prepare:

Grill the chicken until cooked, steam broccoli until tender, and cook brown rice according to package instructions.

350 kcal 50 g carbohydrates
30 g protein 7 g fat

Dinner

Baked salmon with sweet potato and peas

- 80 g salmon
- 100 g sweet potato
- 50 g peas

How to Prepare:

Bake salmon at 180°C for 15 minutes, boil sweet potato until soft, and steam peas.

400 kcal 45 g carbohydrates
25 g protein 15 g fat

System Architecture

